Memo to the CEO - MediScan

From: Sikharam Satya Sri Harsha Phani, ML engineer. Re: Rohit's model, the rebuild, and why "99% accuracy" is the wrong target.

What the 98.4% really was

It cannot be verified: there is no code, split record or weights. What we can show is why any single number from this data deserves distrust: the validation set had 16 images (a perfect score there is compatible with 79% true accuracy); the test set had a different disease mix from training (62.5% vs 74% pneumonia); 15 of 624 test images have a near-duplicate in training; and image size and shape alone separate pneumonia from normal images with ROC-AUC about 0.9, so a model can score well without looking at lungs. The pilot - about 7 in 10 healthy children flagged - is what such a model does on different equipment.

What the rebuilt model achieves

On a locked test set of 837 children's X-rays, split by patient: ROC-AUC 0.989, sensitivity 94.9% and specificity 95.6% at the operating point chosen by cost (a missed pneumonia counts 20x a false alarm). Per 1,000 children at the camp (10% sick): about 95 of 100 sick children are flagged, about 5 are missed and about 40 healthy children are referred unnecessarily; 3 in 10 referrals are healthy. It runs in about 130 ms per image on a laptop CPU (limit 200 ms) and the file is 1.05 MB (limit 25 MB). Both operating requirements are met.

Per 1,000 children at the camp (cost = 20 per missed child + 1 per false alarm).

scenario	sick_missed	healthy_flagged	accuracy	sensitivity	PPV	cost
Always say NORMAL (does nothing)	100	0	90.0%	0.0%	n/a	2000
Pilot model: ~70% of healthy flagged	0	630	37.0%	100.0%	13.7%	630
Hypothetical '99% accuracy', all 10 errors = missed cases	10	0	99.0%	90.0%	100.0%	200
Hypothetical '99% accuracy', all 10 errors = false alarms	0	10	99.0%	100.0%	90.9%	10
ArogyaNet @ cost-optimal threshold 0.515 (locked test)	5.1	39.8	95.5%	94.9%	70.4%	141.3

Why "99% accuracy" is the wrong target

At the camp only about 1 child in 10 is sick. A tool that flags nobody is already 90% accurate and misses every sick child. A tool at 99% accuracy makes 10 errors per 1,000 children: if all are missed cases the cost is 200, if all are false alarms the cost is 10 - a 20-fold difference that accuracy does not show. Our model's accuracy is lower than 99% (95.5%) yet its cost (141) beats the 'all missed' 99% tool. Accuracy also moves with who is screened: the same model scores differently at 73% and 10% prevalence.

Ask for this instead: a minimum sensitivity (for example 95%, judged by the lower end of a 95% confidence interval), a maximum share of healthy children referred, and the PPV and NPV at 9:1 prevalence - measured on patient-disjoint data from other hospitals and, before roll-out, on the camp's own devices.

What it must not be used for

- Diagnosis, or as a reason to send home a child who has symptoms.
- Adults, non-AP views, or X-ray machines and hospitals it has not been re-tested on: on an external hospital the AUC fell from 0.989 to 0.899 and healthy-child specificity from 96% to 73%.
- Any use without a clinician reviewing the "LOW CONFIDENCE" and "PNEUMONIA" results.
- Photos of screens, other images, or non-X-rays: the tool rejects most of these, but this is a safeguard, not a guarantee.

Your three instructions: (1) evidence on what Rohit did wrong - delivered (technical report, Sections 2-3); (2) 99% accuracy - rejected, for the reasons above; (3) 200 ms and 25 MB - met (about 130 ms, 1.05 MB).